BSIDES KC 2026

Threat Hunting Safari

An Intelligence-Driven Hunt for APT33

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SCYTHE

```
$ ./hunt --target apt33
[*] Loading intel report...
[*] Scanning telemetry...
[+] Threats detected.
```

Whoami

Lead Detection Engineer & Deputy of SCYTHE Labs

- Blue Teamer
- DCO Analyst
- Threat Hunter
- Incident Responder

@1qazCasey

/tyler-j-casey

Agenda

01 **What is Threat Hunting?**
Methodologies: Intelligence, Behavioral, Entity-based

02 **Know Your Prey: APT33**
Peach Sandstorm & the Tickler campaign

03 **The Hunt Process**
Hypothesis → Collect → Analyze → Investigate → Refine

04 **The Safari: Live Hunt**
5 hunts through real telemetry in Splunk

05 **Kill Chain & Takeaways**
Reconstructing the full attack flow

PART 1

What is Threat Hunting?

Intelligence-Based

- Driven by CTI reports & advisories
- Hunt based on known adversary TTPs
- Proactive: validate before alerts fire

Behavioral-Based

- Driven by anomalies & deviations
- New service on a single endpoint?
- CMD popups every 2 hours?

Entity-Based

- Focus on high-value assets
- Domain Controllers, file servers
- CEO's endpoint, jump boxes

Today's focus: Intelligence-Based — but watch for behavioral & entity instincts during the hunt

Intelligence-Based Hunting

THE SCENARIO

Your intel team reads Microsoft's August 2024 report: Peach Sandstorm is deploying new custom malware called Tickler, targeting defense and satellite sectors in long-running intelligence gathering operations. They want to validate whether these procedures have been used in your environment.

Read Report
CTI / advisories

→

Extract TTPs
Procedure-level detail

→

Build Hypotheses
What should we see?

→

Hunt
Query telemetry

Don't just read reports — operationalize them.

State-Based Changes

The breadcrumbs adversaries leave behind on the endpoint

Process Creation	PowerShell, CMD from unusual parents	`Sysmon EID 1`
File Creation	Tools dropped to Downloads\APT33\	`Sysmon EID 11`
Registry Modification	HKCU Run key for persistence	`Sysmon EID 13`
Archive Creation	Compress-Archive for data staging	`Sysmon EID 8 / 39`
Network Connection	AnyDesk outbound C2	`Sysmon EID 3`

These are what we hunt for, changes the adversary makes to the endpoint that leave artifacts in our telemetry.

PART 3

The Threat Hunting Process

1

Hypothesis

What are we trying to find? What do we think happened?

▶

2

Collect Data

Gather telemetry. No logs = no proof.

▶

3

Analyze

Look for abnormalities. If found, investigate.

▶

4

Investigate

Associated behaviors? Processes? Communications?

▶

5

Refine

Document, report, educate, shore up gaps.

This cycle is iterative — each finding generates the next hypothesis

PART 2

Know Your Prey: APT33

ALIASES

Peach Sandstorm | Elfin | Refined Kitten | HOLMIUM

- Iranian state-sponsored, linked to IRGC
- Active since at least 2013
- Primary mission: espionage & intelligence gathering
- Has demonstrated destructive capability (wiper malware)

TARGET SECTORS

- Aerospace & Aviation
- Defense & Satellite
- Energy & Petrochemical
- Oil & Gas
- Government

Our Tooling & Lab Setup

HOST-BASED TELEMETRY

Sysmon EID 1	Process Creation
Sysmon EID 3	Network Connection
Sysmon EID 11	File Create
Sysmon EID 13	Registry Value Set
WinEvent 4104	PowerShell Script Block
WinEvent 4688	Process Create (Security)

DETECTION & TOOLS

Splunk
SIEM / log analysis

Sigma Rules
Translated to Splunk SPL

SCYTHE
Adversary emulation platform

Sysmon
Enhanced Windows logging

Real telemetry from adversary emulation — not synthetic data

PART 4

The Safari Begins

Starting from a single CTI report | Pivoting off each artifact we find

```
$ ./safari --intel apt33 --artifact tickler.pdf.exe
```

The Tickler Campaign

Peach Sandstorm — August 2024

Multi-stage backdoor	Tickler drops decoy PDF, executes second-stage DLL (sold.dll)
Double file extensions	tickler.pdf.exe disguises executable as PDF
Registry Run key persistence	Batch script sets Run key named 'SharePoint'
Credential harvesting	LaZagne extracts stored credentials from the endpoint
AD enumeration	ADExplorer snapshots Active Directory for offline analysis
RMM tool deployment	AnyDesk for hands-on-keyboard access
Data staging & exfiltration	Compress-Archive packages data, exfil over C2

Source: Microsoft Threat Intelligence, August 2024

From CTI to Hypothesis

FROM THE MICROSOFT REPORT

"The first sample was contained in an archive file named Network Security.zip alongside benign PDF files used as decoy documents. The archive file contained: YAHSAT NETWORK_INFRASTRUCTURE_SECURITY_GUIDE_20240421.pdf.exe - the Tickler malware."

▼

OUR HYPOTHESIS

If APT33 has operated in our environment, we should find a binary with a double file extension (.pdf.exe) written to disk, a file masquerading as a legitimate document. If present, that binary likely performed follow-on actions: persistence, credential theft, and tooling deployment.

Our approach:

1. Search for double-extension files using a Sigma rule against Sysmon EID 11 (File Create)
2. If we get a hit, pivot to the host and process tree to understand the full scope

https://www.microsoft.com/en-us/security/blog/2024/08/28/peach-sandstorm-deploys-new-custom-tickler-malware-in-long-running-intelligence-gathering-operations/

THE INITIAL HIT In Splunk — Double Extension Detection

Splunk Query — Sigma: Suspicious Double Extension Files (Sysmon EID 11)

```
index=main host=EC2AMAZ-SRGIV5B EventCode=11 TargetFilename="*.pdf.exe"
```

```
index=main host="EC2AMAZ-SRGIV5B"
(TargetFilename IN ("*.exe", "*.iso", "*.rar", "*.zip") TargetFilename IN ("*.doc.*", "*.docx.*", "*.gif.*", "
    *.jpeg.*", "*.jpg.*", "*.mp3.*", "*.mp4.*", "*.pdf.*", "*.png.*", "*.ppt.*", "*.pptx.*", "*.svg.*", "*.xls.*",
    "*.xlsx.*")) OR TargetFilename IN ("*.rar.exe", "*.zip.exe")
| stats values(TargetFilename) by Image
```

Last 4 hours

✓ **4 events** (4/20/26 4:38:00.000 PM to 4/20/26 8:38:14.000 PM) | No Event Sampling | Job | Verbose Mode

Events (4) | Patterns | **Statistics (1)** | Visualization

20 Per Page | Format | Preview

Image	values(TargetFilename)
C:\Users\Administrator.SCYTHE\campaigns\a1.exe	C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._tickler.pdf.exe C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\tickler.pdf.exe

THE INITIAL HIT What We Found

WHAT WE FOUND

File:	tickler.pdf.exe
Full Path:	C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\tickler.pdf.exe
Host:	EC2AMAZ-SRGIV5B.scythe.lab
Source:	WinEventLog:Microsoft-Windows-Sysmon/Operational
EventCode:	11 (File Create)

WHY THIS MATTERS

This matches the Microsoft report exactly — a binary with a double file extension masquerading as a PDF.
The file path (Downloads\APT33\vfs\) is not a standard location — this is a user-writable directory, exactly where adversaries stage tools.
We now have three anchors: **the file, the host, and the path.**

PIVOT → We have a hit. Now we zoom out: filter ALL Sysmon events to this host and expand the time range. What else happened on this endpoint? What process created this file?

SCOPING THE ENDPOINT In Splunk — EventCode Breakdown

Splunk Query — What types of activity occurred on this host?

```
host=EC2AMAZ-SRGIV5B source="WinEventLog:..Sysmon.." | stats count by EventCode TaskCategory | sort -count
```

```
index=main host="EC2AMAZ-SRGIV5B" source="*sysmon*"
| stats count by EventCode TaskCategory | sort -count
```

Date time range

✓ **3,648 events** (4/20/26 7:47:32.000 PM to 4/20/26 7:57:32.001 PM) No Event Sampling

Job Verbose Mode

Events (3,648) Patterns **Statistics (10)** Visualization

20 Per Page Format Preview

EventCode	TaskCategory	count
11	File created (rule: FileCreate)	2532
3	Network connection detected (rule: NetworkConnect)	776
2	File creation time changed (rule: FileCreateTime)	170
1	Process Create (rule: ProcessCreate)	137
22	Dns query (rule: DnsQuery)	16
5	Process terminated (rule: ProcessTerminate)	8
13	Registry value set (rule: RegistryEvent)	5
7	Image loaded (rule: ImageLoad)	2
10	Process accessed (rule: ProcessAccess)	1
12	Registry object added or deleted (rule: RegistryEvent)	1

SCOPING THE ENDPOINT What the Data Tells Us

EVERY EVENTCODE IS AN INVESTIGATION LEAD

EID	Category	Count	What this tells us
1	Process Creation	137	← What commands ran?
11	File Create	2532	← What was dropped to disk?
3	Network Connection	776	← Who did it talk to?
13	Registry Value Set	5	← Persistence?
10	Process Access	1	← Credential theft? LSASS?
22	DNS Query	16	← What domains resolved?

PIVOT → 137 process creation events, let's start there. This is the backbone of any host-based hunt.

HUNT 1 In Splunk — Process Creation (EID 1)

Splunk Query — Build the process tree from parent/child relationships

```
host=EC2AMAZ-SRGIV5B EventCode=1 | stats values(CommandLine) by ParentImage Image | sort ParentImage
```

```
index=main host="EC2AMAZ-SRGIV5B" source="*sysmon*" EventCode=1 (Image IN ("*\a1.exe","*\tickler.pdf.exe") OR ParentImage IN ("*\a1.exe","*\tickler.pdf.exe"))
| stats values(CommandLine) by Image ParentImage
| table ParentImage Image values(CommandLine)
```

Date time range

✓ **9 events** (4/20/26 7:47:32.000 PM to 4/20/26 7:57:32.001 PM) No Event Sampling Job Verbose Mode

Events (9) Patterns **Statistics (7)** Visualization

20 Per Page Format Preview

ParentImage	Image	values(CommandLine)
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\tickler.pdf.exe	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --single-argume C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\NETWORK_INFRASTRUCTURE_SECUF
C:\Users\Administrator.SCYTHE\campaigns\a1.exe	C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe	"C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe" all
C:\Users\Administrator.SCYTHE\campaigns\a1.exe	C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\ADExplorer.exe	"C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\ADExplorer.exe" -snapshot S domain_var_snapshot.dat -accepteula
C:\Users\Administrator.SCYTHE\campaigns\a1.exe	C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\tickler.pdf.exe	"C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\tickler.pdf.exe"
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\tickler.pdf.exe	C:\Windows\SysWOW64\cmd.exe	cmd.exe /c "C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\SharePoint.bat"
C:\Users\Administrator.SCYTHE\campaigns\a1.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell -c (& $env:userprofile\downloads\apt33\vfs\anydesk.exe --start) powershell -c (Get-Process *anydesk* | stop-process)
C:\Users\Administrator.SCYTHE\campaigns\a1.exe	C:\Windows\System32\cmd.exe	cmd /c dir C:\ cmd.exe /c "net use \\EC2-JFHQ02\Z "SecureP@ss11!!" /u:SCYTHE\administrator"

HUNT 1 The Execution Chain — One Process Rules Them All

```
Reconstructed Process Tree from Sysmon EID 1

a1.exe  (C2 implant — the root of EVERYTHING below)
  ├─ whoami                                  ← identity
  ├─ cmd /c dir C:\                           ← file recon
  ├─ cmd /c dir ...\Documents                ← file recon
  ├─ net use                                 ← share enum
  ├─ tickler.pdf.exe                         ← MALWARE (from CTI)
  │   └─ cmd.exe /c SharePoint.bat              ← persistence script
  │       └─ reg add ...\Run /v SharePoint      ← PERSISTENCE
  ├─ lazagne.exe all                         ← CRED THEFT
  │   ├─ reg save hklm\sam ...                 ← hive dump
  │   └─ reg save hklm\security ...             ← hive dump
  ├─ anydesk.exe --start                     ← BACKUP C2 (RMM)
  ├─ ADExplorer.exe -snapshot ...             ← AD RECON
  └─ net use \\Host\Share /u:DOMAIN\User     ← LAT MOVEMENT
```

Sigma: Suspicious CMD/POSH Parent Image | Suspicious LOLBAS Discovery | Suspicious Process Location

KEY FINDING → a1.exe is the root parent for ALL activity: recon, malware deployment, persistence, credential theft, RMM, AD recon, and lateral movement. Every branch of this tree is a hunt we'll trace in the following slides.

HUNT 1 The Discovery Cascade — What the Adversary Learned

Zooming in: Before deploying tools, a1.exe ran a burst of recon. Here's what the adversary was learning:

Command	What the adversary learns	ATT&CK
`whoami`	Who am I? (Administrator.SCYTHE)	T1033
`cmd /c dir ...\Documents`	What documents does this user have?	T1083
`cmd /c dir C:\`	What's on the root drive?	T1083
`net use`	Any mapped network drives?	T1049
`Get-CimInstance ... AV`	What AV/EDR is watching me?	T1518.001
`Get-DnsClientServerAddress`	What's the DNS infrastructure?	T1016
`services --all`	What services are running?	T1007
`arp`	Who else is on this network segment?	T1016

PIVOT → Textbook recon: identity, files, network, AV, creds — all in rapid succession from one process. Now let's see what TOOLS were staged next (Sysmon EID 11).

HUNT 2 In Splunk — File Creation (EID 11)

Splunk Query — What files did the process tree write to disk?

```
host=EC2AMAZ-SRGIV5B EventCode=11 TargetFilename="*\\Downloads\\APT33\\*" | table _time Image TargetFilename
```

```
index=main host="EC2AMAZ-SRGIV5B" source="*sysmon*"
EventCode=11 Image="*\a1.exe"
| stats values(TargetFilename) by Image
```

✓ **22 events** (4/20/26 7:47:32.000 PM to 4/20/26 7:57:32.001 PM) | No Event Sampling ▾ | Job ▾

Events (22) | Patterns | **Statistics (1)** | Visualization

20 Per Page ▾ | Format | Preview ▾

Image	values(TargetFilename)
C:\Users\Administrator.SCYTHE\campaigns\a1.exe	C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._.DS_Store C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._ADExplorer.exe C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._AnyDesk.exe C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._NETWORK_INFRASTRUCTURE_SECURITY_GUIDE_20240421.pdf C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._Policy Guide- April 2024.pdf C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._SharePoint.bat C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._sold.dll C:\Users\Administrator.SCYTHE\Downloads\APT33__MACOSX\vfs\._tickler.pdf.exe C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne_results.txt C:\Users\Administrator.SCYTHE\Downloads\APT33\results.zip C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\.DS_Store C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\ADExplorer.exe C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\NETWORK_INFRASTRUCTURE_SECURITY_GUIDE_20240421.pdf C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\Policy Guide- April 2024.pdf C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\SharePoint.bat C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\sold.dll C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\tickler.pdf.exe

HUNT 2 The Adversary Toolkit on Disk

FILES IN C:\Users\...\Downloads\APT33\

peaches.zip	ARCHIVE	Downloaded via C2, contains the full toolkit
tickler.pdf.exe	MALWARE	Double extension — the Tickler backdoor from the intel report
sold.dll	2ND STAGE	DLL loaded by Tickler for persistence actions
SharePoint.bat	PERSIST	Batch script that sets a Registry Run key
lazagne.exe	CRED TOOL	Open-source credential harvester (browsers, OS, etc.)
AnyDesk.exe	RMM	Legitimate remote access tool — but why is it here?
ADExplorer.exe	AD RECON	Sysinternals tool for offline AD snapshot
lazagne_results.txt	OUTPUT	Credential dump output from lazagne
results.zip	STAGED	Compressed creds — staged for exfiltration
domain_var_snapshot.dat	AD DATA	Full Active Directory schema captured offline

Sigma: Suspicious Double Extension Files | Suspicious Archive Creation | Suspicious Process Location

PIVOT → We've inventoried the toolkit. Were these tools actually EXECUTED? Let's trace tickler.pdf.exe first, did it set persistence? Then lazagne, did it steal credentials?

HUNT 3 In Splunk — Persistence (EID 1 + EID 13)

Splunk Queries — Did tickler.pdf.exe execute? Was a Registry Run key set?

```
EventCode=1 (Image="*\reg.exe)
EventCode=13 TargetObject="*\\CurrentVersion\\Run\\*"  | table _time Image TargetObject Details
```

```
index=main host="EC2AMAZ-SRGIV5B" source="*sysmon*"
EventCode=1 Image="*\reg.exe"
| stats values(CommandLine) by Image ParentImage
```

Date time range

✓ **4 events** (4/20/26 7:47:32.000 PM to 4/20/26 7:57:32.001 PM) No Event Sampling Job Verbose Mode

Events (4) Patterns **Statistics (2)** Visualization

20 Per Page Format Preview

Image	ParentImage	values(CommandLine)
C:\Windows\SysWOW64\reg.exe	C:\Windows\SysWOW64\cmd.exe	reg add "HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run" /v "SharePoint" /t REG_SZ /d "SharePoint.exe" /f
C:\Windows\System32\reg.exe	C:\Windows\System32\cmd.exe	reg.exe save hklm\sam c:\users\admini~1.scy\appdata\local\temp\kudljvfxvw reg.exe save hklm\security c:\users\admini~1.scy\appdata\local\temp\lmgzfady reg.exe save hklm\system c:\users\admini~1.scy\appdata\local\temp\qzbdbbfcbet

```
index=main host="EC2AMAZ-SRGIV5B" source="*sysmon*"
EventCode=13 TargetObject="*\\CurrentVersion\\Run\\*"
| stats values(TargetObject) by Image Details User
```

Date time range

✓ **1 event** (4/20/26 7:47:32.000 PM to 4/20/26 7:57:32.001 PM) No Event Sampling Job Verbose Mode

Events (1) Patterns **Statistics (2)** Visualization

20 Per Page Format Preview

Image	Details	User	values(TargetObject)
C:\Windows\SysWOW64\reg.exe	SharePoint.exe	NOT_TRANSLATED	HKU\S-1-5-21-1322239560-270275548-3257742585-500\Software\Microsoft\Windows\CurrentVersion\Run\SharePoint
C:\Windows\SysWOW64\reg.exe	SharePoint.exe	SCYTHE\Administrator	HKU\S-1-5-21-1322239560-270275548-3257742585-500\Software\Microsoft\Windows\CurrentVersion\Run\SharePoint

TC1

HUNT 3 In Splunk — Persistence (PowerShell - EID 4103)

Splunk Queries — We saw sold.dll being created, but what is it doing?

```
nputObject"; value="Starting execution sequence..."
nputObject"; value="Step 1: Checking monitor size..."
nputObject"; value="Monitor Resolution: 1920 x 1080 pixels"
nputObject"; value="Virtual Screen Size: 1920 x 1080 pixels"
nputObject"; value=""
nputObject"; value="Step 2: Opening PDF file..."
nputObject"; value="PDF file opened successfully: C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\NETWORK_INFRAS

nputObject"; value=""
nputObject"; value="Step 3: Waiting 60 seconds..."
nputObject"; value="Waiting for 60 seconds.............................................................Adding regis

nputObject"; value="The operation completed successfully."
nputObject"; value=""
nputObject"; value="Done!"
nputObject"; value=" Done!"
nputObject"; value=""
nputObject"; value="Step 4: Executing DLL..."
```

```
LogName=Microsoft-Windows-PowerShell/Operational
EventCode=4103
EventType=4
ComputerName=EC2AMAZ-SRGIV5B.scythe.lab
User=NOT_TRANSLATED
Sid=S-1-5-21-1322239560-270275548-3257742585-500
SidType=0
SourceName=Microsoft-Windows-PowerShell
Type=Information
RecordNumber=31773225
Keywords=None
TaskCategory=Executing Pipeline
OpCode=To be used when operation is just executing a method
Message=CommandInvocation(Out-String): "Out-String"
ParameterBinding(Out-String): name="InputObject"; value="Expanded PDF path: C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\NETWORK_INFRASTRUCTURE_SE
CURITY_GUIDE_20240421.pdf"
ParameterBinding(Out-String): name="InputObject"; value="Expanded DLL path: C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\sold.dll"
ParameterBinding(Out-String): name="InputObject"; value=""
ParameterBinding(Out-String): name="InputObject"; value="Starting execution sequence..."
ParameterBinding(Out-String): name="InputObject"; value="Step 1: Checking monitor size..."
```

Slide 23

TC1 Maybe expand this w/ a timeline view around this time and show that sold.dll execution around the same time
Tyler Casey, 2026-04-21T22:28:13.485

TC1

HUNT 3 The Persistence Chain — Confirmed

EXECUTION CHAIN (validated from Sysmon EID 1 + EID 13)

Step 1: a1.exe spawns tickler.pdf.exe	The C2 implant executes the Tickler malware from the intel report
▼	
Step 2: tickler.pdf.exe loads sold.dll	Second-stage DLL handles the persistence setup
▼	
Step 3: sold.dll runs: cmd.exe /c SharePoint.bat	Batch script executes the actual persistence command
▼	
Step 4: reg add HKCU\...\Run /v "SharePoint" /d "SharePoint.exe"	Run key set — named "SharePoint" to blend in as legitimate

Registry confirmed: HKCU\...\Run → "SharePoint" = "SharePoint.exe"

Sigma: Set Run Key Registry Value | Double Extension (Process Create)

CONFIRMED → Persistence set. Matches the intel report exactly. Now: the adversary also ran lazagne.exe — let's investigate credential theft.

Slide 24

TC1 Need to prove step 3
Tyler Casey, 2026-04-21T22:30:02.734

TC1 0 Old note:

cmd.exe /c SharePoint.bat
Batch script executes the actual persistence command
Tyler Casey, 2026-04-21T22:33:22.930

HUNT 3 In Splunk — Credential Harvesting (EID 1 + EID 10)

Splunk Queries — Did lazagne execute? Did anything access LSASS?

```
EventCode=1 (Image="*lazagne*”)
EventCode=10 TargetImage="*lsass.exe" | table _time SourceImage TargetImage GrantedAccess
```

```
index=main host="EC2AMAZ-SRGIV5B" (Image="*\lazagne.exe" OR ParentImage="*\lazagne.exe")
| stats values(CommandLine) by Image ParentImage
```

Date time range

✓ **35 events** (4/20/26 7:43:39.000 PM to 4/20/26 8:03:39.001 PM) No Event Sampling

Job Verbose Mode

Events (35) Patterns **Statistics (4)** Visualization

20 Per Page Format Preview

Image	ParentImage	values(CommandLine)
C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe	C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe	"C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe" all
C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe	C:\Users\Administrator.SCYTHE\campaigns\a1.exe	"C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe" all
C:\Windows\System32\cmd.exe	C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe	cmd.exe /c "reg.exe save hklm\sam c:\users\admini~1.scy\appdata\local\temp\kudljvfxvw" cmd.exe /c "reg.exe save hklm\security c:\users\admini~1.scy\appdata\local\temp\lmgzfady" cmd.exe /c "reg.exe save hklm\system c:\users\admini~1.scy\appdata\local\temp\qzbdbbfcbet"
C:\Windows\System32\inetsrv\appcmd.exe	C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe	C:\Windows\System32\inetsrv\appcmd.exe list apppool C:\Windows\System32\inetsrv\appcmd.exe list apppool .NET v4.5 /text:* C:\Windows\System32\inetsrv\appcmd.exe list apppool .NET v4.5 Classic /text:* C:\Windows\System32\inetsrv\appcmd.exe list apppool DefaultAppPool /text:*

```
index=main host="EC2AMAZ-SRGIV5B" source="*sysmon*"
EventCode=10 TargetImage="*lsass.exe"
| stats values(SourceImage) by TargetImage GrantedAccess
```

✓ **1 event** (4/20/26 7:47:32.000 PM to 4/20/26 7:57:32.001 PM) No Event Sampling

Job

Events (1) Patterns **Statistics (1)** Visualization

20 Per Page Format Preview

TargetImage	GrantedAccess	values(SourceImage)
C:\Windows\system32\lsass.exe	0x1FFFFF	C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne.exe

HUNT 3 In Splunk — Credential Harvesting (EID 1 + EID 10)

Splunk Queries — Lazagne was run, did they stage the files for exfil?

```
04/20/2026 07:52:17 PM
LogName=Microsoft-Windows-PowerShell/Operational
EventCode=4103
EventType=4
ComputerName=EC2AMAZ-SRGIV5B.scythe.lab
User=NOT_TRANSLATED
Sid=S-1-5-21-1322239560-270275548-3257742585-500
SidType=0
SourceName=Microsoft-Windows-PowerShell
Type=Information
RecordNumber=31773096
Keywords=None
TaskCategory=Executing Pipeline
OpCode=To be used when operation is just executing a method
Message=CommandInvocation(Compress-Archive): "Compress-Archive"
ParameterBinding(Compress-Archive): name="Path"; value="C:\Users\Administrator.SCYTHE\Downloads\APT33\lazagne_results.txt"
ParameterBinding(Compress-Archive): name="CompressionLevel"; value="Optimal"
ParameterBinding(Compress-Archive): name="DestinationPath"; value="C:\Users\Administrator.SCYTHE\Downloads\APT33\results.zip"
ParameterBinding(Compress-Archive): name="Update"; value="False"
ParameterBinding(Compress-Archive): name="Force"; value="False"
```

TC1

HUNT 3 What lazagne Did — Credentials Stolen & Staged

LAZAGNE EXECUTION

```
EID 1: a1.exe spawns lazagne.exe all

Under the hood, lazagne:
  └─ reg save hklm\sam ...
  └─ reg save hklm\system ...
  └─ reg save hklm\security ...
  (dumps SAM/SYSTEM/SECURITY hives)

EID 10: lazagne.exe → lsass.exe
  GrantedAccess: 0x1FFFFF
  = PROCESS_ALL_ACCESS (FULL)
  LSASS holds logged-in user creds
```

STAGING FOR EXFILTRATION

```
Output written:
  lazagne_results.txt
  (passwords from browsers, WiFi,
   OS credential stores, etc.)

Immediately staged:
  Compress-Archive → results.zip
  (packaged for exfiltration)

What this means:
  • Adversary has VALID credentials
  • Can authenticate as real users
```

Sigma: HackTool - LaZagne Execution | Suspicious Archive Creation

PIVOT → Credentials stolen and staged. The adversary now has valid creds + persistence. Next: did they deploy AnyDesk for remote access? Did they snapshot AD?

Slide 27

TC1 Prove compress archive, not yet seen

Tyler Casey, 2026-04-21T22:44:47.090

HUNT 4 In Splunk — AnyDesk

```
Splunk Queries — Were the tools from Hunt 2 actually executed?
(EventCode=1 OR EventCode=3 OR EventCode=22) Image="*anydesk*"
```

```
index=main host="EC2AMAZ-SRGIV5B" EventCode IN (1, 3, 22) Image="*anydesk*"
| table Image CommandLine DestinationIp QueryName
```

Date time range

✓ **7 events** (4/20/26 7:43:39.000 PM to 4/20/26 8:03:39.001 PM) No Event Sampling

Job Verbose Mode

Events (7) Patterns **Statistics (7)** Visualization

20 Per Page Format Preview

Image	CommandLine	DestinationIp	QueryName
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe			relay-e24ffb83.net.anydesk.com
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe		143.244.61.217	
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe		186.233.187.24	
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe			relay-31ee9317.net.anydesk.com
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe	"C:\Users\Administrator.SCYTHE\downloads\apt33\vfs\anydesk.exe" --local-control		
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe	"C:\Users\Administrator.SCYTHE\downloads\apt33\vfs\anydesk.exe" --local-service		
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe	"C:\Users\Administrator.SCYTHE\downloads\apt33\vfs\anydesk.exe" --start		

HUNT 4 AnyDesk — Alternative C2 Channel Established

EXECUTION (EID 1)

Process: anydesk.exe --start

Path: ...\Downloads\APT33\vfs\

Parent: a1.exe (C2 implant)

Why this is suspicious:

- AnyDesk normally installs to Program Files
- Running from Downloads\APT33\vfs\ = adversary toolkit
- Launched by C2 implant, not by a user

NETWORK (EID 3 + EID 22)

Outbound connections:

143.244.61.217:443

186.233.187.24:443

DNS queries:

relay-*.net.anydesk.com

Result:

Backup C2 channel established

Hands-on-keyboard access ready

Sigma: RMM AnyDesk Execution | AnyDesk from Non-standard Folder | RAS Non-standard Folder

WHY RMM? Redundancy, Lateral Movement & Blends in.

TC1

Slide 29

TC1 Change this to redundant c2 as well as lat movement w/ “known good” software

Tyler Casey, 2026-04-22T16:09:16.931

TC1

HUNT 4 In Splunk — AD Reconnaissance (EID 1 + EID 3)

Splunk Queries — Were the tools from Hunt 2 actually executed?

```
(EventCode=1 OR EventCode=3 OR EventCode=22) Image="*adexplorer*"
```

```
index=main host="EC2AMAZ-SRGIV5B" EventCode IN (1, 3, 22) Image="*ADExplorer*"
| table Image CommandLine DestinationIp QueryName
```

Date time range

✓ **11 events** (4/20/26 7:43:39.000 PM to 4/20/26 8:03:39.001 PM) No Event Sampling Job Verbose Mode

Events (11) Patterns **Statistics (11)** Visualization

20 Per Page Format Preview

Image	CommandLine	DestinationIp	QueryName
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe		172.31.28.55	
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe		172.31.28.55	
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe		172.31.28.55	
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe		172.31.28.55	
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe		172.31.28.55	
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe		172.31.28.55	
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe		172.31.28.55	
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe		172.31.28.55	
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe			EC2AMAZ-JFOCHEQ.scythe.lab
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe	"C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\ADExplorer.exe" -snapshot SCYTHE domain_var_snapshot.dat -accepteula		
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\ADExplorer.exe	"C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\ADExplorer.exe" -snapshot SCYTHE domain_var_snapshot.dat -accepteula		

Slide 30

TC1 modify splunk query comment at top

Tyler Casey, 2026-04-22T16:16:34.636

HUNT 4 In Splunk — Lateral Movement (net use)

```
Splunk Queries — Were the tools from Hunt 2 actually executed?
(EventCode=1 OR EventCode=3 OR EventCode=22) Image="*net*"
```

TC1

```
index=main host="EC2AMAZ-SRGIV5B" EventCode IN (1) Image="*\net.exe"
| stats values(CommandLine) by Image ParentImage
```

✓ **2 events** (4/20/26 7:43:39.000 PM to 4/20/26 8:03:39.001 PM) No Event Sampling ▾ Job ▾

Events (2) Patterns **Statistics (2)** Visualization

20 Per Page ▾ Format Preview ▾

Image	ParentImage	values(CommandLine)
C:\Windows\System32\net.exe	C:\Users\Administrator.SCYTHE\campaigns\a1.exe	net use
C:\Windows\System32\net.exe	C:\Windows\System32\cmd.exe	net use \\EC2-JFHQ02\Z "SecureP@ss11!!" /u:SCYTHE\administrator

Slide 31

TC1 modify

Tyler Casey, 2026-04-22T16:17:33.681

HUNT 4 AD Recon & Lateral Movement — Crown Jewels Targeted

AD RECONNAISSANCE (EID 1 + EID 3)

```
ADExplorer.exe -snapshot "SCYTHE"
  domain_var_snapshot.dat
```

LDAP connection → 172.31.28.55:389
(Domain Controller)

What was captured offline:

- All user accounts & groups
- Trust relationships
- Full AD schema

LATERAL MOVEMENT (EID 1)

```
net use \\EC2-JFHQ02\Z
  “SecureP@ss11!!" /u:SCYTHE\administrator
```

The connection:

Creds from Hunt 3 (lazagne) →
Used here to mount remote shares

Even if this fails:

- Adversary is trying to spread

Sigma: AD Snapshot via ADExplorer

ENTITY-BASED THINKING: ADExplorer targeting the Domain Controller is entity-based hunting in action. The adversary is going after your crown jewels. If you're not watching AD activity on your DCs, you're missing the biggest moves.

FINAL PIVOT → The adversary has creds, persistence, backup C2, AD intel, and tried to spread. Let's look at the full network picture and tie it all together.

HUNT 5 Net Comms — C2, Exfil & DNS

Splunk Queries — What communications (internal / external) did this endpoint make?

```
(EventCode IN 3,22) Image IN ("*\a1.exe", "*\anydesk.exe", "*\adexplorer.exe")
| stats values(QueryName) values(DestinationIp) by Image
```

```
index=main host=EC2AMAZ-SRGIV5B EventCode IN (3,22)
Image IN ("*\powershell.exe", "*\cmd.exe", "*\a1.exe", "*\anydesk.exe", "*\ADExplorer.exe")
| stats values(QueryName) values(DestinationIp) by Image
```

Date time range

✓ **287 events** (4/20/26 7:43:39.000 PM to 4/20/26 8:03:39.001 PM) No Event Sampling Job Verbose Mode

Events (287) Patterns **Statistics (3)** Visualization

20 Per Page Format Preview

Image	values(QueryName)	values(DestinationIp)
C:\Users\ADMINI~1.SCY\AppData\Local\Temp\ADExplorer.exe	EC2AMAZ-JFOCHEQ.scythe.lab	172.31.28.55
C:\Users\Administrator.SCYTHE\Downloads\APT33\vfs\AnyDesk.exe	relay-31ee9317.net.anydesk.com relay-e24ffb83.net.anydesk.com	143.244.61.217 186.233.187.24
C:\Users\Administrator.SCYTHE\campaigns\a1.exe		13.58.116.24

HUNT 5 Net Commsn — C2, Exfil & DNS

Final spoke: What external communications did this endpoint make? Where did the stolen data go?

ALL EXTERNAL CONNECTIONS (EID 3)

Process	Destination	Port	Purpose	Context
a1.exe	13.58.116.24	443	Primary C2	*All commands + exfil*
AnyDesk.exe	143.244.61.217	443	AnyDesk relay	*Backup C2 channel*
AnyDesk.exe	186.233.187.24	443	AnyDesk relay	*Redundant access*
ADExplorer.exe	172.31.28.55	389	LDAP (DC)	*AD schema dump*

KEY DNS QUERIES (EID 22)

training.stage.unicorncroft.com	*C2 staging domain — IOC for threat intel sharing*
relay-*.net.anydesk.com	*AnyDesk relay infrastructure — backup C2*
EC2AMAZ-JFOCHEQ.scythe.lab	*Domain Controller FQDN — resolved by ADExplorer*

SAFARI COMPLETE → From one double-extension file in a CTI report, we've reconstructed the entire kill chain. Let's put it all together.

PART 5

Full Kill Chain Reconstructed

C2 Implant & Discovery
T1059
T1082, T1083

▶

Tool Staging
T1105
T1036.007

▶

Persistence & Cred Harvest
T1547.001
T1003, T1555

▶

RMM & AD Recon
T1219
T1087.002

▶

Staging & Exfiltration
T1560.001
T1041

HOW WE GOT HERE — PIVOT BY PIVOT

01 **CTI Report:** Microsoft blog → double-extension file

02 **Initial Hit:** Sigma fires on tickler.pdf.exe (EID 11)

03 **Scope Host:** Expand to all events on host

04 **Process Tree:** a1.exe = C2 root of all activity

05 **Recon Cascade:** whoami, dir, AV check, mimikatz

06 **Toolkit Staged:** peaches.zip → 10 files in APT33\

07 **Persistence:** tickler → sold.dll → bat → Run key

08 **Creds Stolen:** lazagne + LSASS 0x1FFFFF

09 **RMM Deployed:** AnyDesk → backup C2 channel

10 **AD Snapshot:** ADExplorer → full AD to DC

11 **Lat Movement:** net use with stolen creds

12 **Exfiltration:** Compress-Archive → exfil over C2

All reconstructed from a single CTI report about a double-extension file.

Sigma Rules Used

Sigma Rule	Detects	EID
Double Extension (File Event)	Files like tickler.pdf.exe	Sysmon 11
Double Extension (Proc Create)	Process with double ext image	Sysmon 1
CMD/POSH Suspicious Parent	CMD/POSH from non-System32 parent	Sysmon 1
LOLBAS Discovery Commands	whoami, net, dir, systeminfo	Sysmon 1
Suspicious Process Location	Exec from Downloads, Temp, AppData	Sysmon 1
Set Run Key Registry Value	Run/RunOnce key modifications	Sysmon 13
HackTool - LaZagne	LaZagne credential harvester	Sysmon 1
AnyDesk Execution	AnyDesk running on endpoint	Sysmon 1
AnyDesk Non-standard Folder	AnyDesk outside Program Files	Sysmon 1
RAS Non-standard Folder	RMM tools from unusual paths	Sysmon 1
AD Snapshot via ADExplorer	ADExplorer -snapshot flag	Sysmon 1
Suspicious Archive Creation	Archive by uncommon process	Sysmon 11

Key Takeaways

- **Start with one artifact from CTI**
  A single double-extension file led us to the entire kill chain
- **Expand outward; investigate every spoke**
  Process trees, files, registry, credentials, network — follow every thread
- **Host-based telemetry is powerful**
  Sysmon + PowerShell logging gave us visibility into the entire operation
- **State-based changes are your breadcrumbs**
  Registry keys, files on disk, process trees, LSASS access, archive creation
- **Sigma rules give you a head start**
  12 rules caught every phase of this attack. Translate and deploy them.
- **You don't need a massive SOC**
  You need curiosity, telemetry, and a process — follow the pivots

Start Your Own Safari

01

Enable Logging

Deploy Sysmon with a solid config. Enable PowerShell Script Block Logging. These two alone gave us the entire kill chain today.

02

Pick One Report

Find a threat intel report relevant to your industry. Extract one artifact. Build a hypothesis. Run the queries. Start small.

03

Build the Habit

Threat hunting is a muscle. The more you practice the pivot chain the faster you get.

Resources: Microsoft Peach Sandstorm blog | MITRE ATT&CK G0064 | Sigma HQ | SCYTHE Community Threats

Thank You

Tyler Casey

SCYTHE

@1qazCasey

/tyler-j-casey

/tylercasey2263

```
$ ./safari --status
[+] Hunt complete.
[+] Kill chain reconstructed.
[*] Happy hunting!
```

Questions?